

Grouping and Emergent Features in Vision: Toward a Theory of Basic Gestalts

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Gestalt phenomena are often so powerful that mere demonstrations can confirm their existence, but Gestalts have proven hard to define and measure. Here we outline a theory of basic Gestalts (TBG) that defines Gestalts as emergent features (EFs). The logic relies on discovering wholes that are more discriminable than are the parts from which they are built. These wholes contain EFs that can act as basic features in human vision. As context is added to a visual stimulus, a hierarchy of EFs appears. Starting with a single dot and adding a second yields the first two potential EFs: the *proximity* (distance) and *orientation* (angle) between the two dots. A third dot introduces two more potential EFs: *symmetry* and *linearity*; a fourth dot produces *surroundedness*. This hierarchy may extend to collinearity, parallelism, closure, and more. We use the magnitude of Configural Superiority Effects to measure the salience of EFs on a common scale, potentially letting us compare the strengths of various grouping principles. TBG appears promising, with our initial experiments establishing and quantifying at least three basic EFs in human vision.

Keywords: vision, Gestalts, emergent features, configural, wholes

When stimuli occur close to one another in space and in time, they may group perceptually into coherent, salient patterns or wholes. Such Gestalts, as they are called, abound in our perceptual world, as when leaves and branches cluster into trees, and when trees merge into forests; when eyes, ears, noses, and mouths configure into faces; when musical notes coalesce into chords and melodies; and when countless dots or pixels blend into a photograph.

The resulting wholes may have properties their component parts lack, such as the identity or expression on a face that is unrecognizable from any one part, or the key in which a melody is played that cannot be deduced from any single note. Understanding how parts combine into perceptual wholes was recognized as a central challenge in perceptual theory nearly 100 years ago (Wertheimer, 1912), but the problem has proven tenacious and remains largely unsolved. The reasons are several but begin with the difficulty in agreeing on definitions for such seemingly fundamental terms as “part,” “whole,” and “object.”

Defining Gestalts

Gestalt phenomena cover a wide turf, one that includes grouping, figure-ground segregation, multistability, and others (for re-

views see Pomerantz & Kubovy, 1986; Kimchi, Behrmann, & Olson, 2003; Peterson, Gillam, & Sedgwick, 2007). The “Gestalt problem,” as it is sometimes called, can be cast in various lights, but the essential concept entails *nonadditivity*, nonlinearity, or emergence. The claim most closely associated with Gestalt psychology is that “the whole is different from the sum of its parts.” That claim is often misstated as “the whole is *greater* than the sum of its parts,” something the early Gestalts never claimed and in fact actively refuted (Pomerantz & Kubovy, 1981). In any case, the notion of nonadditivity is correctly implied by either phrasing.

Beyond nonadditivity, *emergence* is central to perceptual Gestalts. Almost all of perception is nonlinear, starting with sensory transduction. In psychophysics, our senses transduce and encode physical energy in a nonlinear (usually compressive) manner such that doubling the physical intensity of a stimulus rarely makes it appear twice as intense to our senses. So in the case of judging the loudness of two simultaneous tones, the whole is less than the sum of its parts. With emergence, however, not only quantitative but *qualitative* differences appear as parts are added, such that wholes take on properties that are novel, unpredictable, even surprising. These wholes thus have properties not reducible to those of their parts (Stephan, 2003), and thus the whole is different from (rather than more than) the sum of its parts. The element of novelty and surprise lies at the core of a Gestalt.

Background

Important work has been carried out since the initial analyses of Gestalt phenomena by Wertheimer (1912, 1923), Koffka (1935), and Köhler (1947). The research of Kanizsa (1979) and others on stimulus factors underlying Gestalts helped shape the field. Innovative analyses by Rock (1983), Hochberg (1957, 1981), Attneave (1971) and others introduced modern experimental methods, a tradition carried on by Garner (1974) as the information processing

This article was published Online First July 4, 2011.

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We thank Stephen Jewell and Alpna Agrawal for their assistance in conducting the experiment reported as well as for their comments on the theory proposed here. We thank Glyn Humphreys, James Townsend, and two anonymous reviewers for their valuable insights.

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approach became dominant in the analysis of visual perception. Clever experimental innovations and theoretical analyses by Palmer (for an overview see Palmer, 1999) and Palmer's work with Rock (e.g., Palmer & Rock, 1994), by Kubovy (1994) and his coworkers (Kubovy & Van den Berg, 2008), and by many others, have sharpened our understanding of how the elusive Gestalt problem can be approached analytically (despite the seeming oxymoron that suggests; Garner, 1981). The approach presented here draws heavily on that base, and particularly on concepts articulated and summarized most clearly by Garner (1974; see also Pomerantz & Garner, 1973).

Present Approach

In this paper, we approach visual Gestalts by attempting to:

1. define them operationally via *emergent features* (EFs), or as entities containing EFs, that are demonstrable from configural superiority effects;
2. synthesize them from the ground up, starting with the simplest parts and then combining these parts incrementally, noting EFs as they appear; and
3. measure them using performance on standard and familiar visual discrimination tasks so the identified principles of grouping can be quantified and compared with one another on a common scale of measurement.

Although Gestalts can be experienced consciously and can be described in perceptual reports, we approach them here through their specific effects on discrimination performance. Discrimination is the foundation of perception because it entails the registration of differences or of change. Without discrimination, we could not detect changes in luminance across space (and so could not even see edges or contours), or detect changes over time (and so could not see motion or flicker), or detect changes in pressure (and so could not hear sound). Without discrimination, our perceptual worlds would be featureless, homogeneous *ganzfelds* devoid of elements, structure, or perceptual organization.

Understanding how we detect differences in seemingly *basic* stimulus attributes such as color, lightness, or location has been a mainstream challenge in the study of visual perception, and it has met with many successes. A more complex challenge is how we discriminate *complex* stimuli, including whole objects or shapes. How, for example, do we tell a circle from a square? This is an easy discrimination task for humans, but explaining how we do it is complicated by the many physical differences between circles and squares, differences that are hard to isolate and control for, making it difficult to determine which ones are critical for discrimination.

The closely related challenge on which this paper will focus involves the ways perception can be affected by context. When we test how well humans distinguish two stimuli—call them A and B—we can also consider what happens when A and B appear in the presence of a context stimulus C, such that the task is now to discriminate AC from BC. As the Gestalt psychologists noted, wholes differ from the sums of their parts. Thus, when stimulus A occurs in the presence of C, novel features may emerge that are missing from A alone or C alone, and AC's emergent features may

differ too from the ones that arise with BC even though the added context, C, is the same in both cases and thus the signal differentiating AC from BC remains the same.

Four Immediate Goals

Our purpose is to examine contextual or Gestalt effects in visual perception in hopes of determining the number and kinds of emergent features (EFs) that arise in simple visual patterns. To anticipate our conclusions, we suggest that: (a) only a handful of basic EFs occur in human visual perception; (b) although rare, these EFs are robust and can readily be demonstrated experimentally; (c) their salience can be ranked on a common scale of measurement, allowing fair comparisons of their importance in processing; and (d) these EFs play as strong a role, perhaps even a stronger role, in perception than do the nominally more "basic" features or parts from which they emerge.

Figure 1 illustrates the core issue and the experimental approach used here. It shows an *odd-quadrant discrimination task* in which four stimuli—three identical and a fourth different—are presented simultaneously in the four quadrants of a display. The left panel of the top row shows the *base display*: three As and one B, where A and B stand for any two stimuli. The perceiver's task is to locate the odd stimulus, that is, to indicate which quadrant contains the one different from the other three (the B in this case) by pressing that odd stimulus on a touch screen (or the correspondingly positioned key on a keyboard—here, the "3" key to indicate that the odd quadrant is in the lower right). We measure the reaction time (RT) and accuracy of those localization responses.

Often, a single disparate item will "pop out" from a field of identical homogeneous-looking distractors, a result usually explained by appealing to differences in what are often called *basic features* between the target and distractors, features such as size, color, orientation, or shape (e.g., Treisman & Gelade, 1980; Wolfe, 1994; Wolfe & Horowitz, 2004). The ease of spotting the odd item depends upon the visibility of the display and a host of other factors, but it is associated strongly with the perceived dissimilarity of the target from the distractors (Duncan & Humphreys, 1989). That dissimilarity is thought to grow with the number of distinctive features differing between A and B and to decline with the number of shared features (e.g., Tversky, 1977).

The top row's center panel in Figure 1 shows the *context display*: four identical quadrants filled with the context stimulus C. This display is of course not actually presented in the odd quadrant task because it contains no odd quadrant. Rather it is used to create the final *composite display*, shown in the right panel, which is simply the superposition of the base and context displays. How will the speed and accuracy of locating the odd quadrant differ between the composite and base displays? The criterion we use to establish the existence of an emergent feature—a Gestalt—is that performance must be significantly better with the composite than with the base display before we claim an EF has appeared. If performance is not better in the composite, there is no evidence that any new feature emerged; but if it is better, something besides the added context itself must account for the improvement, given that all the added context elements are the same.

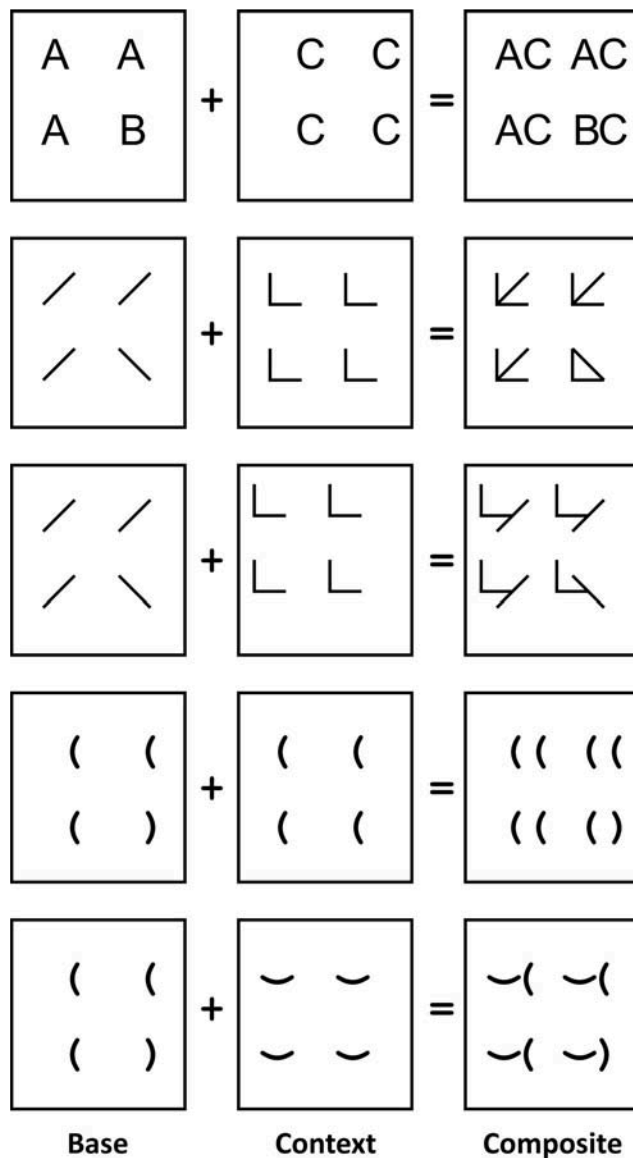


Figure 1. (adapted from Pomerantz et al., 1977). Top row shows a schematic odd-quadrant discrimination task. Participants only see base and composite displays, not context alone. A, B, and C are symbols standing for any stimulus component. The same base stimuli produce CSEs (rows 2, 4) or CIEs (rows 3, 5) depending on the context added, showing that the EF does not appear with just any context but instead depends on context-specific interactions.

A model of perception that excludes Gestalt factors would predict that adding identical context would make it no easier to spot the odd quadrant, because the identical context adds no discriminative information. Stated differently, adding identical contexts should increase the number of shared features but (barring new, emergent features) not the number of distinctive features separating the odd quadrant from the other three. Most models predict such context will *hurt* and cause inferior performance—longer RTs and higher error rates—in the composite display, for at least four interrelated reasons. Namely, the identical context elements might:

1. dilute the differences (increase the similarity) among the quadrants, making the odd quadrant look more like the other three;
2. slow performance by adding to total processing load (more stimulus to process);
3. provide more attention-capturing display items, diverting attention away from the target elements so that the wrong information is processed;
4. mask the target elements through sensory interactions with them (e.g., inhibition); or otherwise impair target processing through crowding (Levi, 2008).

In most cases in which different stimuli are substituted for A, B, and C, this prediction proves correct: performance is worse in the composite display (right panel) of Row 1, Figure 1 than in the base (left). With a small number of stimuli, however, the reverse occurs. Sometimes in fact performance is not just slightly but is dramatically faster and more accurate in the composite than the base display. This result is called a *configural superiority effect* (CSE; Pomerantz, Sager, & Stoever, 1977). Rows 2 and 4 of Figure 1 show two examples: one uses base stimuli that are straight lines differing in orientation, and the other uses curved lines differing in direction of curvature. When such CSEs occur, they arise in spite of the four factors above, thus making our method inherently conservative (as discussed below).

To recap our logic: Because the same context element, C, is added to each quadrant to create the composite display (such as the L added to the diagonal lines), the context by itself adds no information to distinguish the odd quadrant. Its presence will usually make the odd quadrant less detectable, for the reasons noted above. If adding the uninformative context C makes locating the odd quadrant *easier* despite these four factors, some other, offsetting factor must be at work, a factor that allows for what has been called “supercapacity” (wherein added processing load appears to boost performance; see Townsend & Wenger, 2004; Eidels, Townsend, & Pomerantz, 2008).

That offsetting factor, we argue, is the EFs that arise when the context elements interact perceptually with the base elements. In the case of Figure 1, the possible EFs distinguishing the triangle from the arrows in the composite display include closure (a triangle is closed while an arrow is open); line terminators (an arrow has three whereas a triangle has none); intersections (a triangle has three V vertices whereas an arrow has a single fork vertex); and pointing (triangles and arrows are seen to point whereas line segments alone do not). The base display, containing only one diagonal line in each quadrant, contains none of these feature differences.

Slight changes in the context elements can reverse this outcome and convert CSEs into CIEs (*configural inferiority effects*), as shown in Figure 1, row 3. Altering the locations of the horizontal and vertical “L” context elements converts the CSE into a CIE, in that RTs and error rates are now higher in the composite than in the base display. This demonstrates that not *any* context will improve performance (as one might expect if CSEs acted simply like anchor effects in absolute judgment). Instead only those contexts that create EF differences between the odd quadrant and the other

three will lead to CSEs. Row 4 shows another CSE based on line curvature rather than on orientation, where the addition of context elements yields such potential EFs as bilateral symmetry, parallelism, and good continuation or a tendency toward closure. Row 5 shows a corresponding CIE with the same context elements rotated 90 degrees.

A stronger argument is that by finding new CSEs, we can identify and characterize these EFs more precisely and so begin to determine the kinds and range of EFs to which the human visual system is responsive. Ultimately we hope this will lead to a better understanding of context effects, nonadditivities, and Gestalt phenomena in general. Were we to achieve a clear, objective, quantifiable measure for Gestalt effects in perception, that would serve the larger goal of understanding perceptual organization.

We have used this odd-quadrant paradigm with a variety of stimuli, including those shown in Figure 2. Of course there are unlimited stimuli that could be tested, because any three stimulus elements can be substituted for A, B, and C in Figure 1, with further variations possible in their spatial arrangements, orientations, and number. If stimuli are chosen at random for A, B, and C (e.g., characters from computer keyboards), the results from the odd-quadrant task will usually reveal a CIE. Below, we review a few of the cases we have uncovered to date where CSEs result as well as the reverse case where an expected CSE fails to appear. We then attempt to create a systematic way of classifying and organizing these effects. Our goal is to establish a set of basic and principle-driven EFs, features that emerge from combinations of elements but that are more salient (at least to human perception) than are the features of the elements from which they emerge.

Figure 2 shows a few of the stimulus sets we have tested looking for CSEs (Portillo and Pomerantz, in preparation). In brief, we have tested arrangements of letters that do or do not form words to look for equivalents to the well-known word superiority effect and word-letter effect (Reicher, 1969; Wheeler, 1970), and in general we have found none. We have looked at arrangements of line segments that do or do not form letters to see if "letterness" constitutes an EF for human perceivers, but our results indicate not. We have looked at arrangement of line segments and forms that yield potential EFs of surfaces, 3D volume (Enns, 1990), and objecthood; here we have found some cases of significant (albeit modest) CSEs, such as the arrangements of pacman-like figures that yield Kanizsa's subjective figures. We do not, however, find the equivalent to the object superiority effect (Weinstein & Harris, 1974) with our odd-quadrant visual discrimination paradigm. By contrast, the CSEs obtained here with the odd-quadrant task are robust, so much so that every participant tested over the years in this lab has shown the effect with the arrow-and-triangles and with the parenthesis-pairs of Figure 1.

In general our work has shown the largest CSEs when the composite figure target quadrant differs from the other quadrants in terms of topology (Chen, 2005), as is the case with the original arrow versus triangle stimuli in second row of Figure 1. These results show that the CSE is not an isolated result appearing just with those arrow-triangle stimuli or with the parenthesis pairs in fourth row of Figure 1; instead it is a more widespread phenomenon that occurs with a variety of stimuli, some of which produce large and robust effects. The stimuli showing CSE are so complex, however—they differ in so many of their potential EFs—that it is difficult to determine which EF(s) are responsible for observed

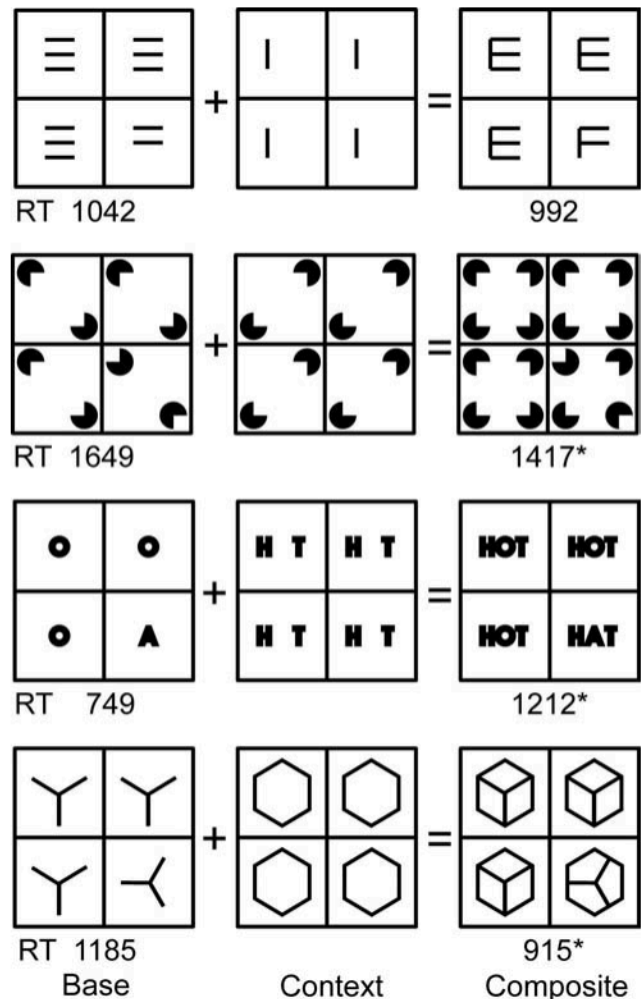


Figure 2. Sample stimuli tested in other experiments (Portillo & Pomerantz, in preparation) looking for further CSEs, including (from top) letters emerging from line segments, subjective shapes emerging from pacman figures, words emerging from letter sequences, and three dimensionality emerging from flat patterns. In some but not all cases, adding identical contexts to to-be-discriminated parts yields better performance in the rightmost composite condition than in the leftmost base condition (i.e., a configural superiority effect).

CSEs. Because of that complication, a new approach is needed, one that simplifies the stimuli in the hope of isolating the effective EFs more clearly.

Purpose of Experiment

Our purpose is to explore what properties of stimuli produce CSEs, along with what kinds fail to do so. Our goal is to identify the fundamental EFs that might underlie all CSEs and show how the stimuli used in previous experiments, as simple as they may be, are still too complex—contain too many confounded differences—to pinpoint the exact EF(s) responsible for the CSEs. Because of that, we argue that a new approach is needed, which we describe below.

The experiment described here uses the method laid out in Figure 1; it differs only in the stimuli employed. Participants performed the odd-quadrant task, locating which one of four stimuli differed from the other three. All four were presented simultaneously, arranged in a square configuration. Participants performed this task both with no added context (leftmost base panels) and with added context (as in the rightmost composite panels of that Figure). In all cases, the odd stimulus was equally likely to appear in any of the four quadrants, and the task was to indicate, as quickly and accurately as possible, which quadrant contained that odd stimulus. We varied which element was odd and which was the background: we displayed both an A in a field of three Bs as well as a B in a field of three As (where A and B stand for any element used). The stimuli remained on the screen until a response was made or time expired, reaction times (RTs) and error rates were recorded, and a brief visual feedback signal was given following error responses.

The key properties contained in every stimulus set are those suggested in Figure 1: in the base display, the A elements must all be identical to each other but different from the B element; the C elements must all be identical to each other (although not necessarily different from A or from B); and the composite display must be the simple superposition of the base and the context display. When identical contexts improve discrimination performance, some offsetting factor must be at work. We posit that this factor is the creation of EFs when A and B are accompanied by the context C.¹

What specifically are these EFs that we claim arise from the perceptual interaction of stimuli with context? Consider Figure 1 Row 2, where Ls are added to positively and negatively sloped diagonal line segments to create arrows and triangles. What differentiating features do the arrows and triangles possess that the diagonals alone lack? At least four candidates suggest themselves:

1. The triangle is a closed form whereas the arrow is open (a topological difference; Chen, 2005; Pomerantz, 2003); line segments themselves are open.
2. The arrow contains three line terminators whereas the triangle has none (also a topological difference); line segments have two.
3. The triangle contains only V intersections (three of them) whereas the arrow has only a single fork intersection; line segments have none.
4. Triangles and arrows appear to point, whereas individual line segments do not.

Did the CSE we found with arrows and triangles result from just one of these EFs, from some combination of them, or from some other factors not yet considered? Given just these stimuli it is impossible to say: the factors are confounded, and dissociating them could be a challenge. Note that in Figure 1 Row 3, where the same diagonals and Ls are combined in a different manner, not all these EFs appear and no CSE arises in their perceptual processing, so there may be something specific about the configurations in Figure 1 that is critical.

Consider next the curves in Figure 1 Row 4. What EFs might be operating here to produce the robust CSE observed? At least three candidates are apparent:

1. One of the pairs has its curved lines parallel whereas the other does not; with an isolate curve, parallelism is not a factor.
2. One of the pairs is mirror symmetric about the vertical axis whereas the other is not; with just one of the curves, there is no such vertical symmetry.
3. One of the pairs has its lines continuing more smoothly into each other suggesting good continuation or closure, whereas the other does not, suggesting either closure or good continuation as the effective EF; for a single curve, good continuation is not a factor.

Once more, there seems to be no way to tease apart these possibilities. Note that if the context is oriented differently as in Figure 1 Row 5, no CSE arises. So again, it is not just any context that yields a CSE; what is needed is one producing a specific perceptual interaction of base display and context that generates an EF.

Herein lies a primary obstacle to understanding the perception of complex stimuli: it is difficult if not impossible to manipulate one property without changing other properties simultaneously. Even with the simple forms used here, confounds abound that make it hard to isolate which factors are most important for human perception. One might think that after all these years we might know why it is so easy for people to tell arrows from triangles. Nonetheless, it appears that even these simple stimuli are too complex—they possess too many featural differences—for us to understand fully how they are discriminated so well compared with any of their components in isolation.

Simplifying the Stimuli

If stimuli like the arrows-and-triangles or the parenthesis pairs are too complex, how might we go about simplifying them? Two possibilities arise: (a) stripping these stimuli down by whittling away at them, element by element, to discover the point at which the CSEs vanish—presumably indicating that the responsible EF has vanished; and (b) taking the reverse tack by beginning with the simplest primitive element (a single pixel, point, or blob) and constructing our stimuli from the ground up, looking for the appearance of EFs at each step along the way.

Figure 3 shows how the first method might work. The top panel shows the same set of curved lines—parentheses—as in Figure 1. In the middle panel the lower half of each curve has been removed.

¹ We have discovered other CSEs in our research, including ones involving topological features; they will be described in a subsequent paper. We are also accumulating evidence that apparent motion may act as an emergent feature by producing a CSE (by presenting context elements earlier in time than the base elements, so as to induce illusory motion). Last, we have suggestive evidence that EFs may occur in auditory perception; for example, when one tone is presented in the context of another tone close in frequency, the resulting auditory beat may produce a CSE and so qualify as an EF.

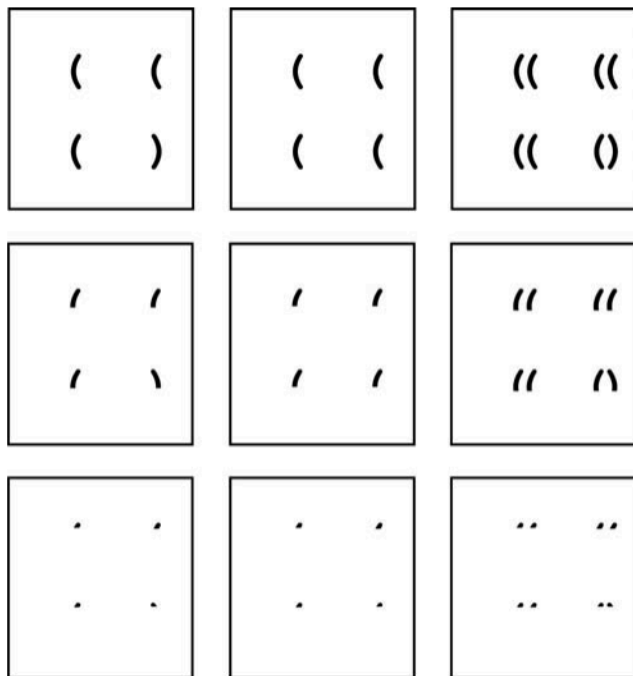


Figure 3. The set of parenthesis pairs from Figure 1 is successively carved away to determine what parts must be present for EFs to exist, as measured by CSEs.

If this stimulus set showed a CSE comparable to that with the lower half intact above, we could conclude that whatever EFs are responsible, the lower half is not essential for their existence. This, in turn, might rule out the tendency toward closure as the critical EF, leaving symmetry and parallelism as remaining contenders. If we continued carving away at these stimuli to where the line segments became too short to retain detectable curvature, we would end up close to the straight-line segments shown the bottom panel. If these stimuli continued to yield an equivalent CSE, we might conclude that curvature itself is irrelevant and that good continuation as well seems like a nonfactor.

If we continue to whittle away, we could erase more pixels and further reduce our stimuli, noticing as we proceed both how potential EFs disappear along with the CSEs they generate. As promising as this method might prove, it suffers from a degree of arbitrariness. Where in the stimulus do we begin the simplification? Is the local approach of removing pixels a good way to simplify a stimulus, or should we attempt to remove more abstract or global properties, as when we took the curves and converted them to straight-line segments in Figure 3?

Rather than grapple with such tangled questions, we opted to take the alternative approach and hunt for EFs by starting from an empty canvas. From that ganzfeld we begin our search by adding pixels, blobs, or perhaps line segments, one at a time, to build complex stimuli and Gestalts from the ground up.

Building Emergent Features From The Ground Up

To create from scratch complex stimuli that contain EFs, we start with the simplest stimuli and increase their complexity incre-

mentally, noting at each step the potential EFs that arise and whether these features lead to the CSEs that are our diagnostic criterion for functional EFs. We begin with a base discrimination between two stimuli and continue by adding identical contexts one step at a time. In this fashion, the base discrimination—the *signal*—remains the same, and therefore any CSEs observed in later composites can be compared on a common scale of measurement. In this fashion, we hope to make fair comparisons among various EFs—that is, among different Gestalt principles of grouping.

Figure 4 shows the basis for our approach, using dots as the starting primitives in the upper row, and line segments in the lower row. Beginning at the upper left, we have what we take to be the simplest stimulus possible (aside from a ganzfeld): a single point (or dot, or blob). The stimulus has no further description than this. Conceptually, even calling it a dot risks suggesting it has a diameter, an area, and properties such as surface color or texture; it has none of these in principle, and it certainly lacks shape or orientation. Points are infinitely small so we must use dots here, but we believe their minimal surface properties do not affect the perception of their configural properties.

The single dot in the upper left panel then possesses only the single property of position. When a second dot is added in the next panel, that new dot too possesses just the property of position. However, two new properties—potential EFs—appear when two dots are present, namely the *proximity* or distance between the two dots, and the *orientation* or angle between them. Proximity and orientation are thus the two most basic or primitive of emergent feature candidates, because they arise from the least amount of stimulus. Our experiments will test whether proximity and orientation are also the most basic in a psychological sense, that is, whether they qualify as EFs by virtue of yielding CSEs.

In the next panel, adding a third dot leaves us with the properties of each of the three individual dots plus the properties of the three pairwise EF candidates of proximity and orientation. In addition, it contributes two new EF candidates unique to triplets of dots: *linearity* (more properly, nonlinearity because two points by definition must be arranged linearly) and *symmetry* (more properly, asymmetry because two points by definition are arranged symmet-

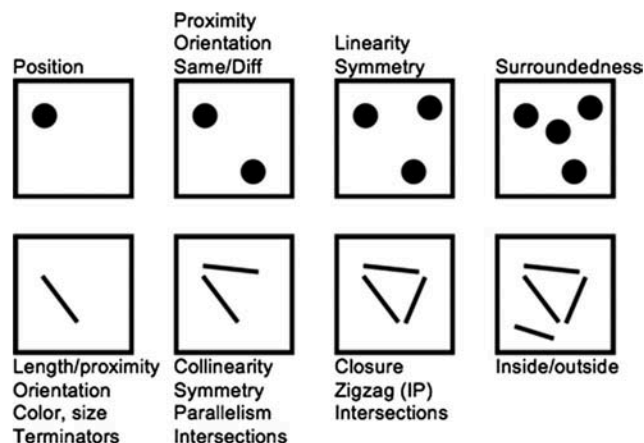


Figure 4. Building complex patterns—and potential emergent features—from the ground up. This is a possible “period chart” of basic EFs arising from configurations of points (top) and lines (bottom).

rically). Finally, adding a fourth dot leaves us with all the individual, pairwise and triplet-wise features but adds one EF candidate unique to configurations of four dots: *surroundedness*, whereby one dot falls fully within the interior or fully in the exterior of the triangular convex hull defined by the first three dots (as distinct from topological surroundedness stemming from grouping by common region; Palmer, 1992).

The second row in Figure 4 takes us through an equivalent progression using short line segments rather than dots as primitives. A single line segment is like a pair of dots in that it has the properties of position, of orientation (angle between endpoints) and of length (proximity of endpoints). It also has the feature of terminators, however—a topological property not possessed by dots or other line-based stimuli such as triangles or quadrilaterals. When a second line segment is added, new properties that emerge include parallelism, symmetry, collinearity, and intersections. Although properties such as parallelism and intersection may be suggested by dot patterns, they reside physically only in patterns containing continuous lines. Again, experiments will tell us if these physically emergent properties also have psychological reality—that is, whether they lead to CSEs and so may also be considered EFs by our criterion.

Adding a third line segment introduces the property of closure (two straight lines cannot themselves form a closed region), a new set of intersections (such as fork and arrow intersections, which require three lines), and a zigzag feature of changing inflection points (e.g., the change from concavity to convexity seen in Ns and Zs). Note again that a property like closure can never be obtained with dot patterns, even though the perceptual organization of dot patterns suggests it. A fourth line segment allows for the topological property of inside versus outside, impossible to achieve with only three line segments or with an infinite number of dots.

The current experiment pursued this approach by looking for the potential EFs of orientation, proximity, symmetry, linearity, and surroundedness. Because of the large number of stimuli, the experiment was split into two parts (1A, 1B) testing different stimuli and different participants.

Method

Participants

All participants were undergraduate students at Rice University meeting psychology course credit requirements through participation. All had normal or corrected-to-normal vision. None of the participants had served previously in similar experiments. Sequences 1 through 3 (Experiment 1A) had 24 participants while Sequence 4 (Experiment 1B) had 20 participants drawn from the same pool as those in 1A.

Stimuli

Thirty-five stimuli (each pairing an A element with a B element) were used, structured from four sequences of dot patterns, with each sequence starting from a common base (see Figures 5–8). The first sequence, shown in Figure 5, starts on the left with the simplest odd quadrant display where each quadrant contains a single dot. In one of the four quadrants, the position of this dot

differs from that in the other three, which are all identical to each other.

Next a second, identical context dot is added to each quadrant. As Figure 5 shows, this context dot can be added in one position to yield the potential EF of orientation or it can be added in a slightly different position to yield the potential EF of proximity.

When the third, identical context dot is added, multiple possibilities arise. To the two-dot display testing for the potential EF of orientation, identical third dots can be added in either of two ways. One way creates the potential EF of symmetry/asymmetry. We will later refer to this stimulus as containing the potential EF of symmetry *derived from orientation* (as opposed to other displays where symmetry is derived from stimuli containing the EF of proximity), and we will designate it as *symmetry/o* (symmetry differences derived from a stimulus already containing orientation differences). Adding the third dot at a slightly different position yields the potential EF of linearity/nonlinearity; we will refer to this as *linearity derived from orientation* and denote it as *linearity/o*. As Figure 5 also shows, with the two-dot patterns testing for proximity, we added only one set of third dots, a set yielding the potential EF of symmetry; we denote this as *symmetry/p*.

Finally, to these three-dot configurations we added a fourth set of identical context dots to create the potential EF of surroundedness; these we denote as *surroundedness/s/o*, *surroundedness/s/p*, and *surroundedness/l/o*. In all, Figure 5 contains eight odd-quadrant displays (*surroundedness/s/p* is absent). Figures 6, 7 and 8 contain nine each, as we explain below. In every case the change that makes the odd quadrant different from the other three quadrants—the “signal”—is identical; it is solely the difference in the location of the same single dot beginning in the base display that generates these potential EFs. Thus, any improvements in discrimination performance moving from left to right in Figure 5 will represent CSEs in which the underlying signal is the same. In this manner, we hope that apples are being compared with apples.

A key feature of this ground-up approach is that it may allow us to spot more clearly what stimulus features are responsible for CSEs. For example, if we find that the symmetric versus asymmetric discrimination (third from left column in Figure 5) yields performance superior to that in the base (leftmost) display, we might be tempted to conclude that symmetry functions as an EF in human vision because it produces a CSE. However, if performance on the immediately earlier orientation display (2nd from left) is as good as on the symmetry display, we would have to reconsider that conclusion, because the superiority seemingly emerging from symmetry may be due entirely to the pairwise orientation differences this display necessarily already contains. For a feature to qualify as an EF in our framework, there must be a CSE arising only when that feature first appears. In a way, this ground-up approach parallels Donders’ subtractive method (Donders, 1868/1969) in that one looks at the incremental change in performance as each additional element is introduced.

Again, the methodology we introduce here is unavoidably conservative. Recall that adding informationless context typically hurts performance for the four reasons noted earlier: dilution of dissimilarities, increased total processing load, attentional diversion, and sensory masking or crowding. Thus, as we continue to add context elements in our search for EFs, any CSEs we observe occur *in spite of* these factors, all of which boost the chances of a CIE. Given this, it is likely that our

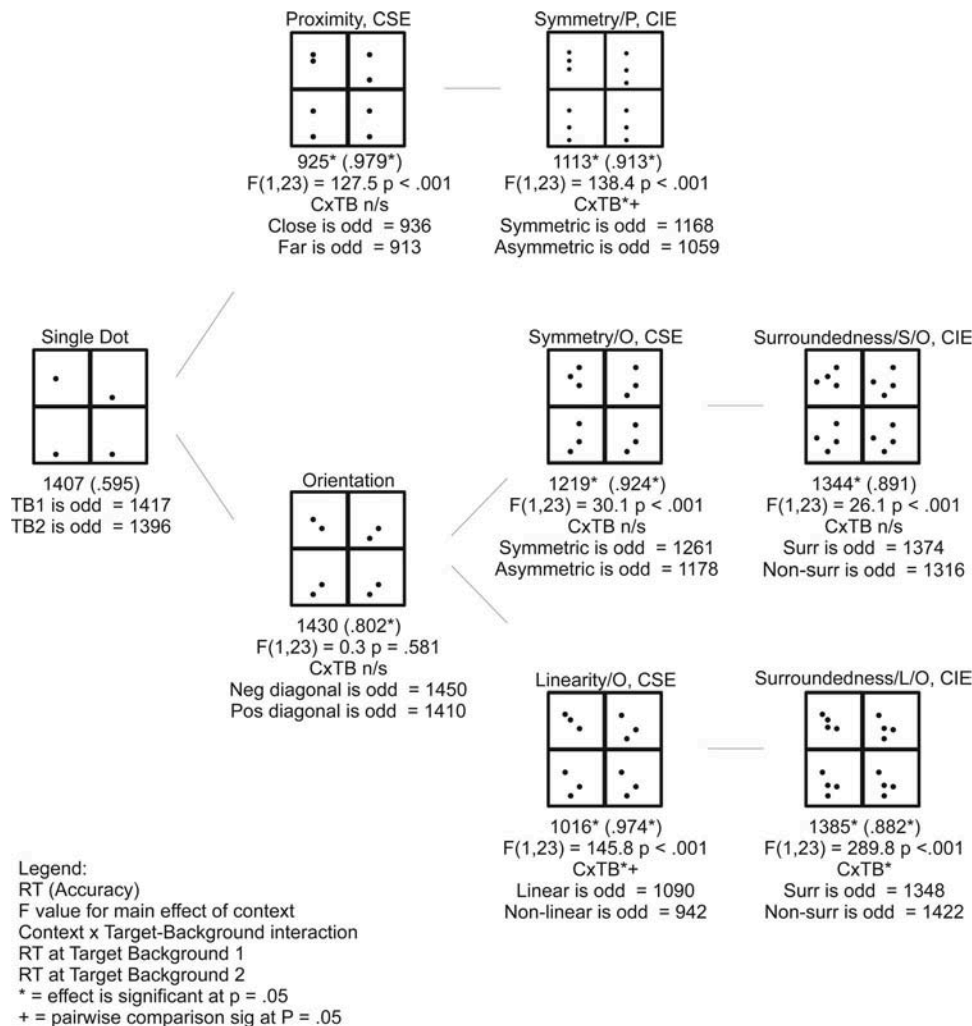


Figure 5. Stimuli and results for Experiment 1A, Sequence 1. In the displays for Symmetry/P and Symmetry/O, the stimulus in top left quadrant is symmetric while those in the remaining quadrants are asymmetric. The F value is for the RT data, and an asterisk (*) indicates $p < .05$.

method will miss EFs that are not large enough to overcome these accumulated liabilities. On the positive side, however, this method is unlikely to produce false alarms, so any EF detected seems likely to be genuine.

In all, the experiment tested the four sequences shown in Figures 5–8. It was divided into two parts, with Experiment 1A testing 26 stimulus sets and 1B testing 9. Each sequence started with a unique base pattern containing just one dot per quadrant and then added sets of context dots to create different potential EFs or to create the same EFs through different dot arrangements. Sequence 1 (in Figure 5) contained eight basic odd-quadrant displays, each of which had eight variations differing in which quadrant was odd and in which element served as target and which as background. Sequences 2, 3, and 4 each had nine basic odd-quadrant displays as shown in Figures 6, 7, and 8.

Experiment 1A. Experiment 1A included three base stimulus pairs containing one dot per quadrant. Adding a second, identical context dot to each quadrant resulted in 2 two-dot stimuli (one

containing the potential EF of orientation and the other of proximity). Adding a third dot led to three subsequent stimuli containing the potential EFs of linearity/o, symmetry/o, and symmetry/p. Lastly, a fourth dot added to the linearity and symmetry stimuli yielded two stimuli containing the potential EFs of surroundedness (surroundedness/s/p, surroundedness/s/o and surroundedness/l/o). This method keeps the signal constant throughout the sequence so that any effect observed in the composite display must be the result of the newly created EF.

Experiment 1B. Experiment 1B introduced one more dot sequence (Sequence 4) in which the potential EF of orientation did not depend on a comparison against a canonical orientation (vertical or horizontal) or on a 90 degree difference between the target and the distractor. This sequence was included to determine whether the results obtained in Experiment 1A generalize to other possible orientation differences.

All stimuli appeared in black on a white background using Corel Draw and were converted to bitmap files. The entire stimulus

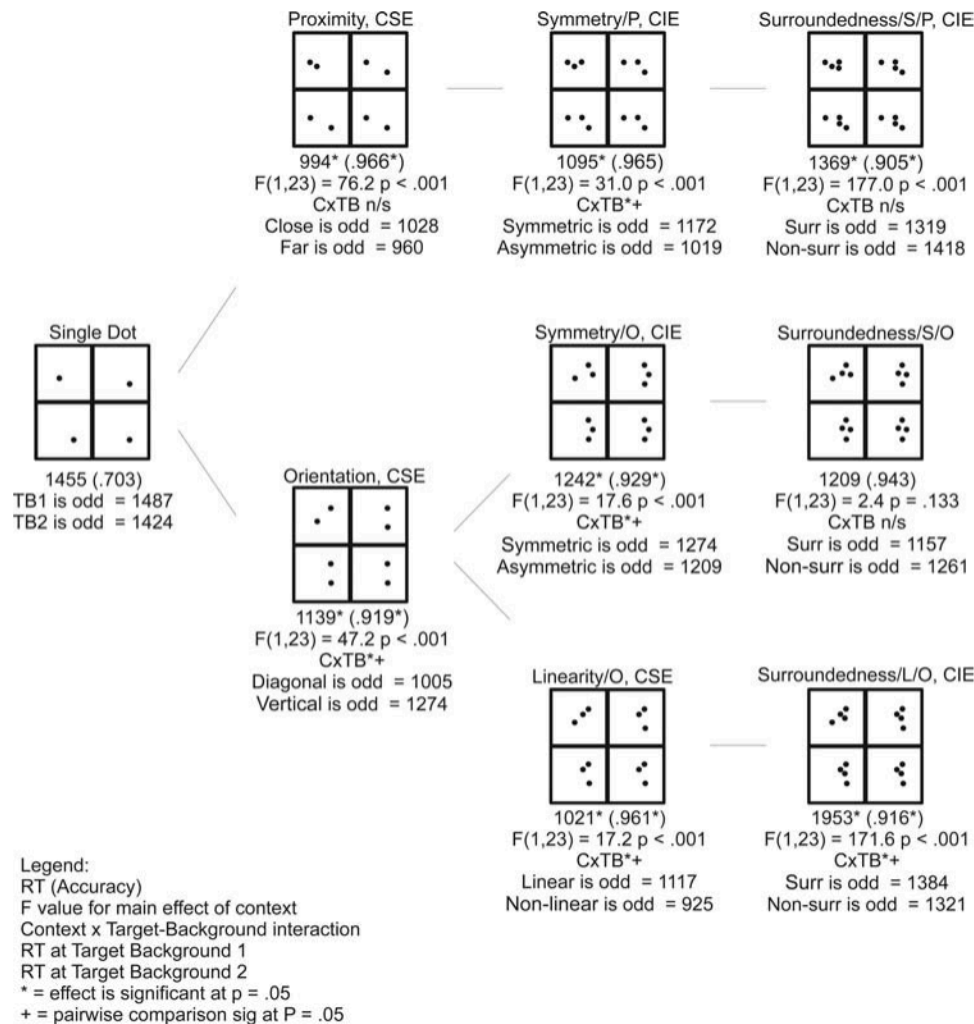


Figure 6. Stimuli and results for Experiment 1A, Sequence 2. In the displays for Symmetry/P and Symmetry/O, the stimulus in top left quadrant is symmetric while those in the remaining quadrants are asymmetric. The F value is for the RT data, and an asterisk (*) indicates $p < .05$.

display encompassing four quadrants measured 17.8 by 17.8 cm, so they subtended a visual angle of ~ 6.6 deg. at a viewing distance of ~ 76 cm. Stimuli were displayed on Sony Trinitron 17 in. GS220 monitors running at 640×480 resolution and controlled by generic PCs running E-Prime software under Windows 98se. The monitors were calibrated to minimize display differences in color and brightness. The stimuli appeared on the screen until a response was made or until 2.5 sec elapsed, whichever was briefer.

Design

The experimental design contained four factors in a $35 \times 2 \times 2 \times 4$ design. First was *Stimulus*: namely, the 35 stimuli just described. Second was *Context*: whether the base stimuli appeared alone (base condition) or with noninformative context present (composite condition). Third was *Target-Background*: which of the two elements served as the target defining the odd quadrant and which served as the background in the other three quadrants

(denoted A in a field of Bs or vice versa). The fourth factor was *Quadrant*, indicating which of the four quadrants contained the odd element (locating that quadrant was the task). Each of the resulting 320 unique displays was presented twice over the course of the experiment. We used a fully repeated measures design, with each S serving in all the cells. All the stimuli appeared once in a first block and again in a second, with context present once and absent once; within each block, stimuli appeared in random order in blocked groups of eight presentations defined by the four quadrants and two target-background combinations of a set. The order in which stimuli were presented was randomized across participants. RT and accuracy were recorded for each presentation.

Procedure

Following a fixation dot in the center of the screen, participants were shown a display that was divided into four quadrants by four gray box outlines, each containing one of the elements from a stimulus. Three quadrants were identical to one other (i.e., they

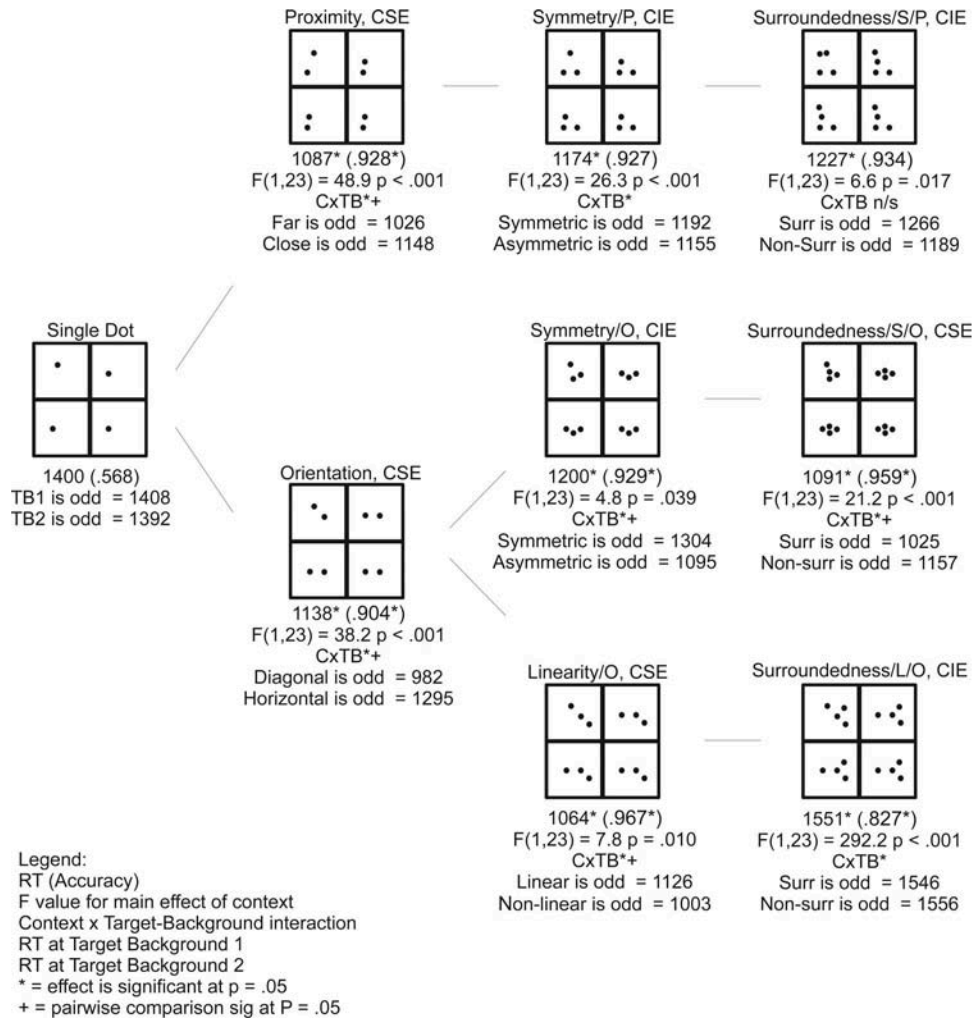


Figure 7. Stimuli and results for Experiment 1A, Sequence 3. In the display for Symmetry/P the top left quadrant of the display is symmetric while the remaining quadrants are asymmetric. The display for Symmetry/O has symmetric dot arrangements in all quadrants, so the discrimination made by the subject was based on the orientation of the axis of symmetry. The F value is for the RT data, and an asterisk (*) indicates $p < .05$.

could be superimposed by simple transposition with perfect registration); a fourth quadrant was always different. Participants were instructed to locate that odd quadrant as quickly as possible by pressing the corresponding key in the number pad of a computer keyboard (7 = northwest quadrant, 9 = northeast, 1 = southwest, and 3 = southeast quadrant). The No. 5 key was used as a home key that participants were required to press before seeing the first stimulus and after every 16 trials, to initiate the next 16.

Each odd quadrant display remained on the screen until a response was made or for a maximum of 2,500 ms, after which the trial was terminated and recorded as "no response." A correct or incorrect response also terminated the display. The participants received visual feedback of "incorrect" on incorrect trials and "no response" on no-response trials. The feedback for incorrect responses did not specify which quadrant had been the correct choice in the previous trial, though. "No response" trials were not used in the RT or error analyses.

The experiment began with a practice block that used as stimuli the pairs $\times$ versus O, A versus B, and red versus green squares. When this practice concluded, the actual trials in the four-factor design followed in two blocks as described above.

Results

Our analyses examined RTs (correct responses only) and error rates across the various stimulus sets. The error data patterns showed no inconsistencies with the RT patterns. RTs below 200 ms in both experiments, and RTs above 2294 ms (Experiment 1A) and 2354 ms (1B), were considered outliers and removed prior to the analysis. Responses beyond the 2.5 sec cutoff were not recorded. Each stimulus was evaluated with a repeated measures ANOVA using the four factors of Context versus No-context, Target-Background combination, Odd quadrant, and Block.

The results of greatest interest were the CSEs and CIEs across the stimulus sets. These results are included alongside the stimuli

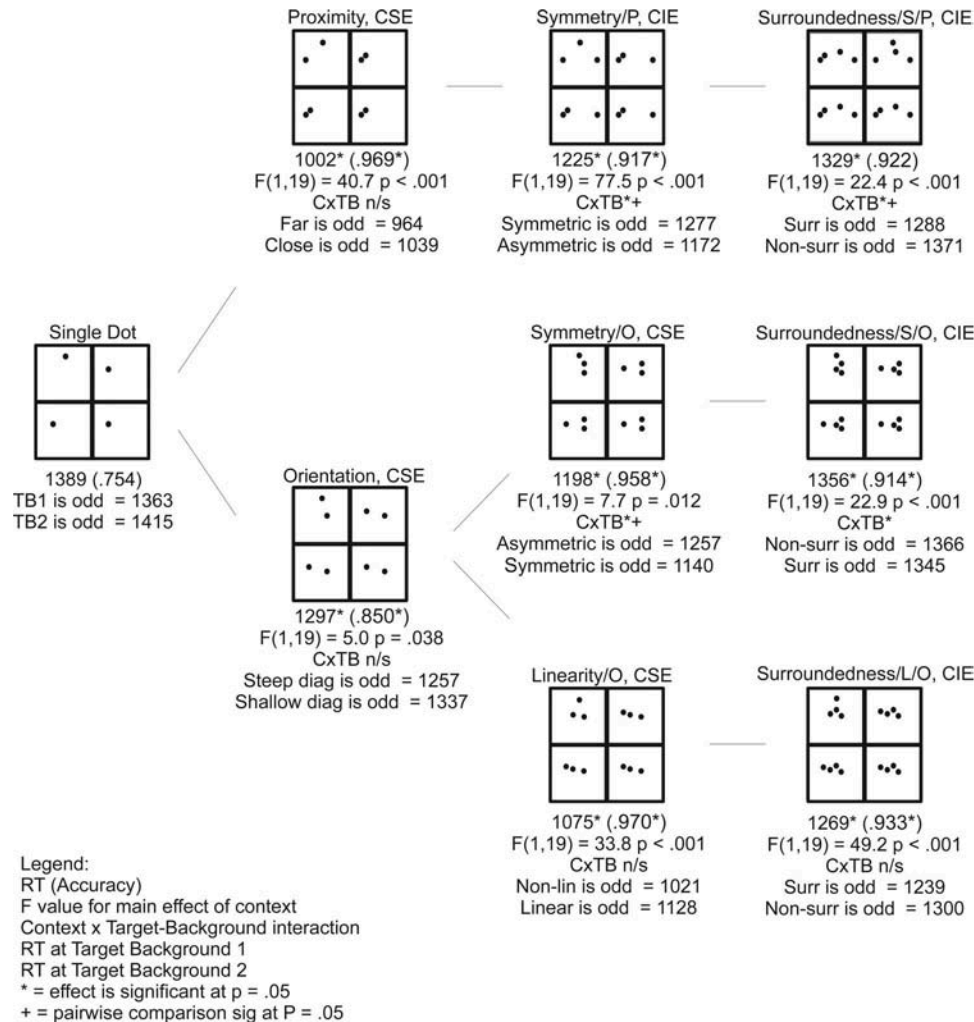


Figure 8. Stimuli and results for Experiment 1B, Sequence 4. In the display for Symmetry/P the top left quadrant of the display is symmetric while the remaining quadrants are asymmetric. The display for Symmetry/O has symmetric dot arrangements in all quadrants, so the discrimination made by the subject was based on the orientation of the axis of symmetry. The *F* value is for the RT data, and an asterisk (*) indicates $p < .05$.

in Figures 5, 6, 7, and 8. Beginning with the stimuli testing proximity as an EF, the results across all four sequences show statistically significant drops in RT from the base condition to the composite condition. In all cases, the improvement in RT exceeds 390 ms. This RT pattern shows evidence that the target is identified more quickly when a difference in proximity emerges from the second dot. Stated differently, Ss are quicker to determine the distance between two dots than the position of either of those two single dots alone in their quadrants.

Orientation produced CSEs in Sequences 2, 3 and 4 (see Figures 6, 7, and 8). The statistically significant RT drop was present when the task required a vertical versus diagonal discrimination, a horizontal versus diagonal discrimination, and a discrimination between two negative slopes of varying angles, as in Sequence 4. Only Sequence 1 failed to show a CSE; it trended toward a CIE, although it did not reach statistical significance. That sequence required discrimination between

dots that represented a negative diagonal and a positive diagonal, always a difficult task for people. In short, Ss are better at determining the orientation (angle) between two dots than the position of either dot.

Symmetry/p (symmetry derived from proximity) produced CIEs in all the sequences. Even though in some instances the RT loss was less than 100 ms, the effect was statistically significant in all cases. Further, Sequences 2 and 3 produced statistically significant CIEs for symmetry/o (symmetry derived from orientation). At first blush, these data seem to indicate that symmetry does not behave as an EF. We should note, however, that the symmetry derived from orientation in Sequences 2 and 3 was derived from diagonal versus vertical/horizontal discriminations, which had already shown great RT gains when compared to their base. Further, symmetry/o produced statistically significant CSEs in Sequences 1 and 4. It is possible, then, that the lack of CSEs may be the result of a floor effect in RT—the previous CSEs had basically made

discriminations so easy that added EFs could not make them any easier. This possibility will be addressed in the discussion.²

Linearity/o (linearity derived from orientation) produced CSEs in all the dot sequences. Along with proximity, linearity shows the most consistent evidence that it may be an EF in human vision. Stated differently, Ss are better at judging whether the slope and intercept between two points A and B matches those between B and C (and thus whether the two dot pairs are collinear) than they are judging the slopes and intercepts associated with single dot pairs.

Surroundedness shows mixed results but in general produced CIEs. In all the cases where surroundedness was derived by adding a dot to symmetry/p or linearity/o stimuli, it produced a statistically significant RT increase from the base to the composite condition. Also, two other cases produced CIEs for surroundedness/s/o (surroundedness derived from symmetry derived from orientation, in Sequences 1 and 4). A trend toward a CSE was seen in surroundedness/s/o (in Sequence 2) but it failed to reach statistical significance. However, the last case of surroundedness/s/o (in Sequence 3) produced a significant CSE.

Search Asymmetries

We also analyzed our results for search asymmetries (Treisman & Gormican, 1988; Treisman & Souther, 1985), under which the search for an A in a field of Bs yields different performance from searching for a B in a field of As. According to Treisman's *feature integration theory*, the appearance of search asymmetries is one diagnostic criterion for calling a feature basic. Because it is easier to search for the presence of a feature than for its absence (Neisser, 1967), such a search asymmetry may help narrow the identity of a basic feature. Alternatively (but often similarly) it is generally easier to search for a stimulus that is like a prototype but containing a small deviation among in a field of stimuli lacking a deviation, so it is easier to search for a circle with a tick mark in a field of circles without ticks than vice versa. In our experiments, for example, participants searched for a dot located high in its box within a field of dots placed low, or vice versa. When an identical second dot was added to the first, subjects looked for a pair of dots oriented on the negative diagonal in a field of dots pairs on the positive diagonal, or vice versa. When an identical third dot was added, they search for a linear triplet of dots in a field of nonlinears (three-dot clusters defining a triangle) or vice versa. This may help us determine whether the EFs arising with each added dot yielded distinct search asymmetries, that is, whether these asymmetries themselves emerged rather than just persisting identically across each sequence. In this way, emergent search asymmetries may provide another diagnostic for EFs.

We first evaluated the interaction between the Target-Background and Stimulus variables to determine whether any search asymmetry when context dots were added differed from search asymmetry in the base before context was added; if so the asymmetry truly emerged, either simply growing stronger, or more interestingly, reversing direction. We followed this test with a pairwise comparison of both levels of the Target-Background variable for each Stimulus.

With the EF of *proximity* arising from a second (context) dot, a statistically significant Context $\times$ Target-Background interaction arose only in Sequence 3. Here, participants were faster finding a

distantly spaced arrangement of dots among closely spaced distractors than vice versa, and the pairwise comparison showed this difference to be statistically significant. The same trend appeared in the other three dot sequences although the Context $\times$ Target-Background interactions were not significant. In all cases, far targets were detected more quickly than close targets.

With the EF of *orientation*, Context $\times$ Target-Background interactions appeared in Sequences 2 and 3, and in both cases searching for a diagonal among horizontal (or vertical) distractors was faster than vice versa. The pairwise comparisons confirmed that these trends were statistically significant in the composite display. Search for a positive diagonal orientation among negative diagonal orientations, and search for an orientation 15° off vertical among orientations 60° off vertical (Sequences 1 and 4, respectively) was faster than their corresponding reverse searches, but neither reached significance in the Context $\times$ Target-Background interaction.

Symmetry/p produced significant Context $\times$ Target-Background interactions in all four sequences. In all cases the form of the interaction was the same: search was faster for asymmetric targets among symmetric distractors than vice versa. The pairwise comparison was significant in confirming this effect in three of the four sequences (Sequence 3 was the sole exception). For *symmetry/o*, this same interaction was significant in Sequences 2, 3, and 4, and the trend was again consistent: in all but one sequence (*symmetry/o* in Sequence 4), search was faster for asymmetric targets among symmetric distractors than vice versa, and the pairwise comparisons were significant whenever the interactions were significant.

Linearity/o also produced consistent search asymmetries. The Context $\times$ Target-Background interaction was significant in all but Sequence 4 (where the RT trend was at least present): participants were faster to locate nonlinear targets among linear distractors than vice versa, and the pairwise analysis showed significant results whenever the interaction was present.

Surroundedness produced mixed Context $\times$ Target-Background interactions. *Surroundedness/s/p* produced an interaction only in Sequence 4, in which the pairwise comparison also proved significant. Targets with surroundedness were located more quickly among distractors without surroundedness than vice versa. *Surroundedness/s/o* produced interactions in Sequences 3 and 4. In both cases search was faster when the target contained surroundedness, but the pairwise comparison was significant only in Sequence 3. *Surroundedness/l/o* produced Context $\times$ Target-Background interactions in Sequences 1, 2 and 3. In two of these cases, Sequences 1 and 3, search was faster when the target showed surroundedness but the pairwise comparison was not significant in neither of these cases. In Sequence 2 both the interaction and pairwise comparisons were significant but the direction of the trend was opposite to that of Sequences 1 and 3, that is, search

² One reviewer points out that a certain EFs (symmetry, but possibly linearity or orientation as well) may be more salient when they appear near one of the borders that framed our quadrants. This seems very plausible. For example, it might be easier to judge the orientation of two dots with a nearby box creating a reference frame. Note, however, that both the borders and the preceding dots were present in the base, so an EF that arises only near a border is still an EF, but in this case an EF that involves the border. We have experiments planned that move these borders further way, or change them otherwise, to determine if they play a critical role in CSEs.

was faster when the target showed nonsurroundedness and the distractors showed surroundedness than the reverse. In sum, there were a number of consistent search asymmetries that may provide supportive evidence for proximity, orientation, and linearity EFs.

Relative Strength of Emergent Features

A key feature of our technique is comparing EFs arising from a common starting point. When comparing proximity with orientation, both have the same single dot as a starting point; and with linearity/o versus symmetry/o both have orientation as a common base in each sequence. The results of these comparisons for each of the four dot sequences are presented in Table 1.

We ran paired *t* tests to look at the difference between the RT gains produced by proximity or orientation compared with the one produced by linearity or symmetry. Our results show that except for the nonsignificant result in Sequence 3, the EF of proximity always produced a larger RT gain than did orientation. Similarly, linearity/o always produced a RT gain larger than symmetry/o. Our technique does not allow us to compare proximity against linearity or other potential EFs that lack the same base. Thus, we start with this partial ordering and hope that in the future we may design more stimuli that would allow for such comparisons.

Discussion

Proximity, Orientation, and Linearity

Evidence from these dot stimuli suggests that proximity and linearity function as EFs in human vision, because each yielded a clear CSE. Further, there is evidence that orientation too serves as an EF. For the first two, all sequences showed CSEs: Proximity produced a superiority effect whether the two dots were arranged in a diagonal or in a vertical orientation; and Linearity produced CSEs independent of the particular distances between the dots or their orientation. Orientation produced CSEs whenever the discrimination was between a negative diagonal and a horizontal, between a positive diagonal and a vertical, or between two negatively sloped dot arrangements. The only exception to orientation's functioning as an EF came in Sequence 1, where orientation appeared as a positive versus negative diagonal discrimination. The results for this particular stimulus, though, might be explained by the oblique effect (Appelle, 1972), which indicates that our visual system is better at discriminating between vertical and

horizontal lines than between diagonal lines. In agreement with this well-known effect, Pomerantz et al. (1977) found slower RTs and CSEs when their stimuli were positive and negative diagonal line segments than vertical and horizontal lines.

Symmetry

We found inconsistent support for symmetry as an EF. When derived from proximity (symmetry/p), it failed to produce a superiority effect but when derived from orientation it produced CSEs in two out of four cases. On the other hand, the search asymmetry analyses show significant effects going in the expected direction in seven of eight comparisons. Some of our distractors were nearly symmetric, and one of our comparisons was between symmetry about a cardinal axis versus a noncardinal axis. So it could be that with other contrasts, our method would better support symmetry as an EF.

Overall in the literature there are mixed signals as to whether symmetry should be considered an important, fundamental visual feature or organizing principle (Wagemans, 1995). Although symmetry (particularly mirror symmetry) is frequently a salient aspect of perceived form, and although some Gestalt demonstrations suggest it can be a factor in perceptual organization, there are also indications that symmetry is less important than it may seem (Pomerantz & Kubovy, 1986). Humans can be good at identifying symmetry along vertical and horizontal axes but with dotted stimuli they are poor along diagonal axes and when symmetry occurs far away from the central axis of the object (Julesz, 1971; Rock, 1983; Wagemans, 1995). Symmetry is readily detected in single patterns but searching for symmetry among multiple patterns can be slow, with positive search slopes (Olivers & van der Helm, 1998). Such poor performance would help explain the CIEs obtained with the dot stimuli presented. Note too that in most of our symmetry stimuli, the target/distractor distinction was defined on the presence or absence of any axis of symmetry but in some cases, target and distractor both contain symmetry so the task became one of discriminating the axis of symmetry of the target from that of the distractors (see symmetry/o in Sequence 3 and 4). However, since both the discrimination of symmetry's presence and of its axis produced both CSEs and CIEs, little can be concluded now. Given the conservative nature of our criteria for defining EFs, and given the considerations mentioned above, it would be premature to rule out symmetry completely. But little can be said now that strongly affirms symmetry as an EF based on our data.

Table 1

Relative Strength of Potential Emergent Features (EFs) in Experiments 1A and 1B. The Mean RT Gain was Calculated by Subtracting the Gain Obtained From Proximity From the Gain (or Loss) Obtained From Orientation. The Same Procedure was Followed for Linearity/o and Symmetry/o

Stimulus sequence	Features	Mean RT gain (ms)	<i>t</i> value	<i>p</i> value
1	Proximity-Orientation	486	13.86	< .001
2	Proximity-Orientation	132	4.87	< .001
3	Proximity-Orientation	43	2.05	0.052 n/s
4	Proximity-Orientation	291	7.44	< .001
1	Linearity/o-Symmetry/o	190	8.54	< .001
2	Linearity/o-Symmetry/o	216	8.02	< .001
3	Linearity/o-Symmetry/o	121	5.19	< .001
4	Linearity/o-Symmetry/o	123	5.86	< .001

Surroundedness

CIEs were obtained for surroundedness in all except Sequences 2 and 3, that is, surroundedness/s/o. Although the conservatism of our method permits us to confirm an EF when a CSE arises, it makes us less able to reject one on the basis of a CIE. As with symmetry, it will be important to test for surroundedness with additional stimuli before ruling it out totally as an EF. One reason is that a related form of surroundedness—topological surroundedness resulting from enclosing elements with a single common region—has been shown to be an effective grouping principle in normal vision (Palmer, 1992). That form of surroundedness has also been shown effective in overcoming the visual extinction found in certain neuropsychological patients whereby an element that they would not normally perceive becomes perceptible if it is grouped with another element in the opposite visual field (Humphreys, 1998).

Because CIEs are useful for ruling out simple anchor effects as explanations for CSEs—they show that not just any anchor or context helps performance—their presence indicates that the CSEs obtained for proximity, orientation, and linearity are not simply a result of using context dots as “reference points;” instead there must be an organization of these dots into new features that are highly salient to the visual system.

Search Asymmetries

Again search asymmetries have been proposed both by Treisman and Souther (1985; Treisman & Gormican, 1988; Wolfe, 2001), as a diagnostic criterion for basic features. The nature and directions of search asymmetries are not yet fully understood, and the approach we offer here does not take a firm stand on what direction a particular asymmetry should take, only that the emergence of asymmetries can support the claim that a potential EF is an actual EF. Our experiments yielded consistent search asymmetries with stimuli that produced CSEs, thus providing independent confirmation that those stimuli contain EFs that act as basic features do. We found search asymmetries in our composite displays, asymmetries that either were absent in the base displays or went in the opposite direction from those base displays. That in turn indicates that the responsible factor was something emerging in the composite displays, namely an EF.

Typically, visual search performance is faster for a diagonal line in a field of verticals than vice versa; for a curved line in a field of straight lines; for an open circle in a field of closed circles; for a circle containing a tic mark (Q) than one without; for a moving stimulus than one stationary; and so forth. If we regard the vertical, the closed, and the symmetric as canonical or prototypical, and if we view the other stimuli as deviating from the prototype by the addition of an extra feature, our present results are consistent: With *proximity*, although we found significantly faster search for distantly spaced dots among closely spaced dots than vice versa in only one of four comparisons, the other three comparisons went in that same direction. This is consistent with the finding that it is easier to detect long lines in a field of shorts than vice versa, and it suggests that distance (remoteness) is the effective feature and that proximity is the absence of distance, so far as the human visual system is concerned. For *orientation*, we find significantly faster search for diagonal targets among verticals or horizontals than vice

versa in two of two comparisons. (On the other two comparisons, search was faster for steeply tilted targets against shallowly tilted distracters than vice versa, and for positive diagonals among negative diagonals than vice versa, but neither proved significant). For *linearity*, we found significantly faster search for nonlinear among linear in three of four comparisons (the fourth went in the same direction); and for *symmetry*, there was faster search for asymmetries among symmetries in seven of eight comparisons (with the eighth going in the same direction). This suggests not only that the dimensions of diagonal versus nondiagonal, linear versus nonlinear, and symmetric versus asymmetric correspond to EFs but that the “feature present” poles of these EFs may be diagonality, nonlinearity, and asymmetry respectively. For *surroundedness*, we find little evidence that it acts as an EF and little basis for predicting a privileged direction or canonical side of the dimension; consistent with that, we find no consistent search asymmetries (in six cases, surrounded prevailed whereas in two, nonsurrounded prevailed). The directions of these search asymmetries are also consistent with our “ground up” method; for example, when a third dot is added to a two-dot display, asymmetry and nonlinearity appear (symmetry and linearity are present by definition in all two-dot displays).

In short, the search asymmetry results provide independent evidence that EFs in human vision act like highly salient basic features—actually more salient than the features of the component parts—and show their own search asymmetries. Although not all the contrasts proved significant, their directions went in the expected directions a total of 18 of 19 times.

In sum, we conclude that proximity, orientation, and linearity function as EFs in human vision. Although there is almost no evidence for it in our data, it is possible that symmetry too is an EF. It failed to produce noteworthy, consistent CSEs here, but the conservatism of our measure suggests we cannot count it out completely. There is only the weakest of evidence that surroundedness acts as an EF. Our analysis on the strength of EFs suggests that proximity is a much stronger EF than orientation, and that even if it is shown subsequently that symmetry can act as an EF, symmetry is much weaker than linearity in grouping.

General Conclusions

Toward a Theory of Basic Gestalts

We began this paper with a review of the Gestalt problem, which has been with us for about 100 years (Wertheimer, 1912). We all seem to know what a Gestalt is—it is a pattern, a configuration, a whole that emerges from a set of parts. At the same time, we seem to have multiple definitions of a Gestalt, when we have any crisp definitions at all.

We reviewed a few veteran configural superiority effects, where wholes are discriminated better than are their component parts, and we introduced a few new ones. With CSEs, people are better (say) at judging the distance or angle between two dots than they are at judging the position of either dot alone, a result that could be surprising given that one straightforward way to compute distance and angle requires starting with the coordinates of the endpoints. Similarly people are better able to judge whether triplets of dots fall along a straight line (rather than forming a triangular pattern) than we would expect given how well they process simple pairs of

dots. Research now underway suggests people are better at judging whether two lines are parallel than they are at judging the orientation of either individual line (Stupina & Pomerantz, 2009), and older work showed people are better at telling arrows from triangles than they are telling apart the single line segment that distinguishes that arrow from the triangle. These results too may be surprising if one believes that we perceive complex objects by integrating their basic components, features, or parts (Pomerantz, 1978). Features such as orientation are often claimed to be basic, and indeed a tilted line does pop out from a field of vertical lines. But our research suggests that parallelism (instead of, or in addition to, orientation) is the basic feature, and that a tilted line pops out from a field of verticals because it introduces the new emergent feature of nonparallelism (Pomerantz & Portillo, in press).

It is often noted, correctly, that human perception is relative. People are poor at judging the absolute brightness of a surface, the size of a circle, or the tilt of an edge. Do the CSEs demonstrated in this paper, and the EFs they imply, simply reconfirm this ancient observation? No—perception may be relative, but not all relations are created equal. Our results show that only a few specific relationships are salient in human vision. If adding context elements per se were helpful (e.g., because they provide anchors), then CSEs would abound. Empirically, however, most added contexts hurt and lead to CIEs. Our explanation is that only contexts yielding EFs will help. CSEs point to those EFs and show us what basic Gestalts—wholes with relationships among parts—exist in the human visual system.

We introduce here a new approach in the form of preliminaries to a theory of basic Gestalts (TBG), which will try to place these results within a unified framework. We view this as an attempt to solve the Gestalt problem, or at least a portion of it, by simplifying the problem into more basic components and defining key terms (Pomerantz, 2001). In truth, the Gestalt problem is probably a set of problems that cannot be solved in a monolithic fashion, but the effort here tries to speak to the core issue of emergence and nonadditivity of parts.

Our preliminary version of TBG entails several claims:

1. Gestalts—wholes that possess properties different from their parts—may be equated with emergent features (EFs)
2. EFs May Be Diagnosed by Configural Superiority Effects (CSEs)
3. the Relative Strength of EFs May Be Scaled by the Magnitude of These CSEs
4. These magnitudes can be compared directly against each other for different EFs when the methodology holds constant the physical signal creating them
5. EFs exist in a hierarchy, such that further EFs appear as new components are added to a stimulus
6. It is possible, at least in some cases, to unbundle the EFs of a complex stimulus by establishing the EFs of progressively simpler stimuli and looking for evidence of additional EFs through further CSEs in the more complex stimuli.

The main claim of TBG is that Gestalts can be equated with an emergent feature (EF), and that EFs in turn can be defined by the appearance of configural superiority effects, themselves rooted in performance on simple discrimination tasks. Thus, we are defining a Gestalt as any relational feature that emerges when parts are juxtaposed in space (and time), so long as that juxtaposition results in a boost in perceptual discriminability. Thus, when we consider a discrimination between two line segments that differ in orientation, and we then add an identical second line segment to each, these new line pairs may be more discriminable from each other than the two original line segments were from each other, because of the emergence of a novel EF. In this case that EF could be parallelism or collinearity (or yet another feature, possibly even symmetry). Whatever the EF may be, the boost in discriminability—the CSE—implies that *some* new feature must have emerged. Otherwise we would have expected the added context to have impaired discriminability.³

This approach lets us sort out would-be Gestalts from the genuine article. One might claim, for example, that any arbitrary pair of stimuli yields a Gestalt or configuration. Taking an intentionally wild example, we might claim that the ratio of the width of one's nose to the height of one's ear constitutes a perceptual Gestalt. For that claim to be substantiated, according to TBG these stimulus elements would have to yield a CSE, which seems unlikely in this instance. In any event, a good theory of Gestalts must have a clear criterion for Gestalthood in order to establish what is *not* a Gestalt.

The method underlying TBG that we have introduced here, by which we build Gestalts from the ground up, appears to allow us not only to identify Gestalts but in some instances to measure those Gestalts on a common scale. In the early days of Gestalt psychology, the “laws” for perceptual organization—grouping, figure-ground segregation, and the like, proliferated with abandon (Pomerantz & Kubovy, 1986), to the point where there were hundreds of them, more laws than there were phenomena those laws were intended to explain. Besides our inability to rule out most of these laws for lack of clear criteria for Gestalthood, we were also unable

³ In reviewing a draft of this article, James Townsend noted that EFs often appear to involve only portions or subsets of the elements of a configuration and not the entire whole. We acknowledge that some Gestalts engage the entire stimulus and not just one of its properties such as linearity while also wondering how one might know that was the case with any particular stimulus. We propose to pursue the current approach with progressively more complex stimuli to see if we can find instances of configuration that cannot be explained by emergent features. In addition, the approach here rests heavily on configural superiority effects—enhancement of discrimination performance—to diagnose grouping and emergent features. There are other forms of evidence—converging operations—that supplement CSEs and make the theory presented here less dependent on a single diagnostic measure. One of those supplements involves the search asymmetries that we present here and show operate with emergent features independently from how they operate on the simpler elements from which those novel features emerge. Similarly, stimuli that produce CSEs have also been shown to yield Garner Interference (a case in which selected attention does not occur to individual elements that group into larger units—see Pomerantz & Garner, 1973; and Kubovy & Pomerantz, 1981). Elements that produce configural superiority also show redundancy losses and supercapacity effects in a redundant targets paradigm (Eidels et al., 2008).

to measure the potency of those left standing on any numerical scale. Thus to this day we have no unit of measurement that tells us whether grouping by proximity is a stronger or weaker principle than grouping by lightness similarity. We hope the method we introduce here is a step in the direction of addressing that issue. It may be that one day we will all measure grouping strength in agree-upon units (perhaps *Wertz*, in honor of Max Wertheimer).

When it can be applied successfully, our method allows us to build stimuli with increasingly more complex features—and thus the potential for more EFs—such that the resulting stimuli always differ from each other physically in exactly the same way they started. As we have seen, when individual dots that differ only in their position on a screen are joined by a second dot, those dot pairs now differ not only in the position of the initial dot but also in the Gestalts of the distance (proximity) between the dots and the angle (orientation) between the dots. The base signal—the physical difference between the stimuli—remains unchanged because the context that is added is the same for each stimulus. When a third dot is added, the potential EFs of symmetry and linearity arise, but again the signal remains the same. By looking at the size of the CSE that results from these new features, we can scale their magnitudes and so have a numerical measure for the potency of each Gestalt—each EF—on a level playing field.

We remain far from knowing how many such basic visual Gestalts exist beyond the three we confirm here. It is possible that the number is finite and manageable, and so we might avoid the proliferation that plagued early Gestalt psychology. It would be noteworthy should their number ultimately turn out to be in the range of the number of phonemes in spoken languages, or the number of geons in recognition by components theory (Biederman, 1987). The preliminary theory sketched here takes no stand on how many EFs might be discovered, however.

In its strongest form, the claim is that grouping consists of the creation of EFs, which in turn are demonstrable not only through direct observation and perceptual reports but through performance indicators such as CSEs. If we ask whether two dots group into a dot pattern, we are basically asking if the perceptual system computes or extracts EFs between the two. With just two dots, that would be their proximity and angle; with three dots their linearity (and possibly symmetry) and so forth. With n elements, we say they all form one group if the visual system computes and processes a feature unique to that ensemble of n (with simple clumping/massing/density counting only as the most crude group possible, if indeed it counts at all). EFs involving subsets of those n are important but irrelevant to whether the specific n elements group. If no such EF is computed, we say those n elements do not group (although again, subsets of them may well group). If we extract and attend to these EFs, that is what it means to group those affected dots. With dot lattices, if we see grouping into rows, that means we are detecting the EFs specific to rows and not those specific to columns; we are processing such possible EFs as the orientation of those rows, the dot and row spacing, density, and symmetry. The prediction would be that observers would be poor in noting those same properties in the unattended columns. Although this view of grouping may be flawed and may fail to capture some aspects of a group, we intend it as a step in the direction of being more explicit and precise in what the term means. It is offered as an alternative to the claim that elements

grouped together merely “belong together” or “complete a figure,” expressions whose meaning is less precise.

In addition, if two items group, that does not necessarily mean they are glued together. Instead, their parts may continue to be as perceptible as they were before any grouping took place (although in some cases components may cease being perceptible, as with *eliminative* properties; see Kubovy & Van Valkenburg, 2001). It also does not necessarily mean that the whole is perceived more quickly or accurately than its parts. The forest does not necessarily hide the trees. Instead, grouping implies that something new has emerged, a feature that is available now that was unavailable before—often a novel and surprising feature.

Limitations of Dots and of Lines

We launched our experiments with dots because the CSEs that had been discovered with parenthesis pairs and with arrows and triangles—those showing the EFs from curved line and tilted line segments—are too complex to analyze. To this day we do not know what makes an arrow so highly discriminable from a triangle that differs from the arrow only by the orientation of its diagonal. Even simple forms like these have too many features to pinpoint which ones perceivers are using. Small numbers of dots, by contrast, are simpler, but evidently they are complex enough to possess Gestalts.

A limitation arising with dot patterns, however, is that they are not able to yield some EFs that may be basic. Consider the property of closure, for example, which is one candidate for an EF that makes a triangle so discriminable from an arrow. One cannot generate closure from a finite number of dots—there will always be gaps between the dots when they are spaced out. Therefore, to get to properties like closure, as well as to other topological properties like the inside-outside distinction—we have to change our basic building blocks from dots to line segments. Some initial experiments using line segments appear promising and are generating a new set of EFs that cannot be captured by simple dots (Stupina & Pomerantz, 2009). One downside of changing from dots to line segments is that we lose our baseline signal—the common physical stimulus difference among the stimuli being discriminated that allows us to place all Gestalts in a single experiment on a common scale of measurement. We will continue to look for ways to skirt that limitation.

It is unclear how far we can go using line segments as the building block for increasing complex configurations, but it may be the case that once that vein is tapped out, we will again have to switch building blocks, moving to the next level of complexity. It is unclear what that next level might be, but following the progression from dots to lines, it would seem one likely candidate would be line pairs (V-intersections) or triangles. The logic here is that two dots define a line, but three dots define a triangle or a pair of lines. That work is still on the horizon but is worth considering at this early stage.

Limitations of Conservatism

There is another limitation to TBG, mentioned several times previously: the conservatism of our method for measuring Gestalts. By our method, to qualify as a Gestalt—an EF—a feature must yield a CSE, that is, a benefit from the addition of

informationless context. As we have seen, that is no easy feat because of the four factors that must be overcome to avoid the opposite effect, the CIE, where context hurts. When identical elements are added to two different stimuli, the resulting composites should have diluted levels of similarity; the added elements can distract attention, they can induce sensory masking or crowding; and they can increase overall processing load. All of these factors will impair performance, not improve it.

In addition to these, another factor may make our method overly conservative: as we lay on additional contexts to create possible higher-order EFs, we require that each potential EF create a CSE over and above any CSEs generated by lower-order EFs. So in Figure 5, to qualify as an EF the symmetry feature emerging with the third dot must generate a CSE over and above that generated by the proximity or orientation feature emerging from the second dot. As we continue to pile context on top of context to generate a hierarchy of EFs as our approach demands, we are increasing the prospects of ceiling or floor effects—of making stimuli so discriminable that they become maximally discriminable—so no additional featural differences can improve discrimination performance.

The result is that our method is almost certainly too conservative. The good news is the low likelihood of our procedure's issuing a false alarm and incorrectly identifying a feature as an EF. The bad news is the near certainty that TBG will fail to detect perfectly good Gestalts, emergent properties that are highly salient to human vision but simply no more salient than other EFs already contained within them. In other words, there may be Gestalts lurking within stimuli that are apparent phenomenologically to perceivers but that simply cannot make their presence known through a CSE because of these factors working against them. For that reason we are reluctant to dismiss symmetry and surroundedness entirely, although our data offer no support.

An ideal correction to this conservatism would be a method to make an independent estimate the total impact of these negative factors. If we knew the extent to which added context produced lateral masking or crowding, for example, we could correct for that when we calculate CSEs, in effect giving features a handicap or extra credit for the challenges they must overcome to yield a CSE. With such a procedure in place, we would then find that features we currently cannot claim as EFs would become proper EFs—and therefore confirmed, basic Gestalts—once this correction is applied.

Have We Identified The Correct Emergent Features?

A longstanding challenge in perceptual research is identifying the effective features by which we perceive stimuli. The challenge originates in the large number of features that objects possess and the difficulty of manipulating those features independently. It is impossible, for example, to alter the size of an object while holding both its shape and its perimeter length constant.

Much research is conducted with rich stimuli like scenes, complex objects, and faces, stimuli that contain dozens if not hundreds of identifiable features. This complexity makes the challenge of identifying the effective features even more daunting. Our ground-up approach to this problem has been twofold: first, to start with the simplest of all stimuli—the dot—where the number of features is minimal; and second, to add elements—more dots—

one at a time, noting any new features that emerge along the way. In the limiting case dots have no spatial extent, color, or texture; the only features they possess are their coordinates in space. So when a second dot is added to the first on a page, we have the *x*, *y* coordinates of each separate dot, but we also have the distance (proximity) and angle (orientation) between them as features that we have shown to be effective for human perception and that are more salient than the *x*, *y* coordinates of either dot.

Is it possible that proximity and orientation are not the effective EFs with two-dot stimuli? Could there be alternative features that are the ones that count? For example, when two dots are brought together, dot density naturally increases, which in turn alters the local luminous flux. If it turned out that luminous flux was the effective feature, that would be valuable information; but it would not alter the theoretical approach offered here, it would change only the specific features the theory identifies.⁴ When we add context to base stimuli, clearly *something* is emerging in the composite. We would need to relabel that novel feature to be sure, but the buildup of CSEs from additional context, and the emergence of new search asymmetries would remain unchanged. There may similarly be an alternative variable that correlates with orientation, although none suggest themselves. The same could be true for alternative features that are correlated with linearity, symmetry, and surroundedness. For example, when three dots form a straight line, it is also true that the distance between the two furthest dots is the sum of the two other interdot distances; so perhaps the latter is the effective emergent feature. Such an explanation would be in harmony with Julesz's (1981) notion of order statistics, although that approach did not bear much fruit in explaining texture segmentation (e.g., Martin & Pomerantz, 1978). Such an explanation would be hard-pressed to explain why adding a third dot makes dot pairs more discriminable only when one of them now forms a linear or symmetric pattern but not otherwise, because in all cases the second order statistics on interdot distances and angles have changed.

What solutions might there be for this problem? First, if the EFs we focus on here are misidentified, those EFs will then fail to generalize to other stimulus sets and so the problem should become apparent as we extend this work into other domains. Second, it is possible too that these alternative features, once identified, could be disentangled experimentally from the ones we identify here. One could try to unconfound dot proximity from luminous flux, for example, by altering the lightness of the stimuli to compensate exactly for those flux changes, as Chen (2005) has done in demonstrating that the effectiveness of topological features cannot be attributed to flux. Or one could test the stimuli at isoluminance. Should such experiments succeed, that would certainly cause us to relabel the effective features responsible for our observed effects but would otherwise leave our explanation of

⁴ We thank an anonymous reviewer for pointing out that when we discover an EF from the addition of Dot 2 to Dot 1, and then find an additional EF from adding Dot 3 to Dots 1 + 2, we must separately test adding Dot 3 to Dot 1 to ensure this second EF does not stem from that pairwise interaction rather than from the triple interaction we claim. It seems unlikely that this effect would have occurred repeatedly in our experiments, but nonetheless we will be conducting the appropriate experiments to rule this out explicitly.

basic Gestalts and emergent features unchanged. Our hope is that once these fundamental EFs—basic Gestalts—have been identified, we can then use them as a starting point for analyzing the configural processes that take place with more complex objects, faces, and scenes—stimuli that are studied intensely today but with little independent control over their effective features.

Limits to TBG

TBG is intended to address some of the main issues in perceptual organization but certainly not all of them. It does not address the fundamental problem of figure-ground segregation, for example, or the phenomenon of multistability found with Necker Cubes and other stimuli. Instead, it focuses on the problem of grouping, that is, of establishing which components of the visual field are linked with which others to form perceptual objects.

Can TBG handle, even in principle, all possible Gestalt grouping principles? An easy answer is “probably not.” Recall that hundreds of Gestalt laws have been proposed over the years. At the same time, some of these “laws” are opaque or cryptic—it is unclear what they mean—and experimental evidence for them is lacking. Helson’s (1933) list included 114 Gestalt laws, and even E. G. Boring’s (1942) short list of 14 included such entries as “Naturalness of form,” “articulation,” “strong and weak forms” (distinct from good and poor forms), and “meaningfulness of forms.” The challenge going forward is for TBG to rephrase as many of these laws as possible into underlying EFs with known geometric underpinnings, and to determine for any remaining laws not recast in these terms whether there is, in fact, any substance behind the claimed law. However this turns out, we should end up with a clearer sense of what these Gestalts might be, whether or not they can be handled by TBG in the final analysis.

In conclusion, we have tried in this paper to take on one of the central problems in perceptual organization, namely defining what we mean by Gestalt and producing a method for quantifying them. Our product at this stage has been a method, one that seems to have promise, and also a set of early Gestalts, perhaps the most basic, that this method has identified: proximity, orientation, symmetry, linearity, and surroundedness.

Should the approach offered here take root and thrive, we will then be able to say with greater precision what it means when we claim that “a whole is different from the sum of its part.” Recast in terms of TBG, this means that when simple stimuli appear in the presence of other contextual stimuli, sometimes powerful new properties called emergent features arise, features that are processed more accurately and more quickly than are the features of the stimuli from which they arise.

This paper has dealt with simple stimuli, starting with mere dot pairs. Ultimately, we want to understand the grouping processes that take place when we view much more complex stimuli like faces and natural scenes. As complex and unwieldy as such stimuli may be for taming in the laboratory, the underlying idea is that they involve a large number of EFs, assembled into a hierarchy of parts whereby more local regions are grouped by the most basic of EFs such as proximity and orientation. More global properties, in turn, are grouped via the kinds of EFs that surface only with more stimulus elements and a higher level of complexity.

When we claim that perception is wholistic, it means we are attending to and processing the EFs, as opposed to processing the

features of the elements from which the EFs arise. When we claim that a complex stimulus like a face is seen as a configuration, it means that we are attending to the EFs that arise when subcomponents of the entire stimulus—say the mouth, the nose, the eyes—come into spatial and temporal proximity. When we say that a melody is heard independently of the pitch of any of its notes, it means we are attending to the EFs—the frequency intervals and timing—by which successive notes are related. When we say that we see a whole before we see its parts, it means that the EFs are processed more quickly than are the features of the parts, the central characteristic of CSEs. This does not mean that the more basic features are not processed at all—that the forest hides the trees—although surely that happens under some circumstances.

There is no reason in principle why Gestalts need be cloaked in the mystery that has surrounded them for so long. There is no reason why they should be seen as soft or fuzzy concepts best left to phenomenal demonstration, and no reason why they cannot be studied as straightforwardly, as psychophysically, as any other stimulus property.

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Received May 14, 2010

Revision received March 17, 2011

Accepted March 27, 2011 ■

